PRINTABLE REFERENCE CHART

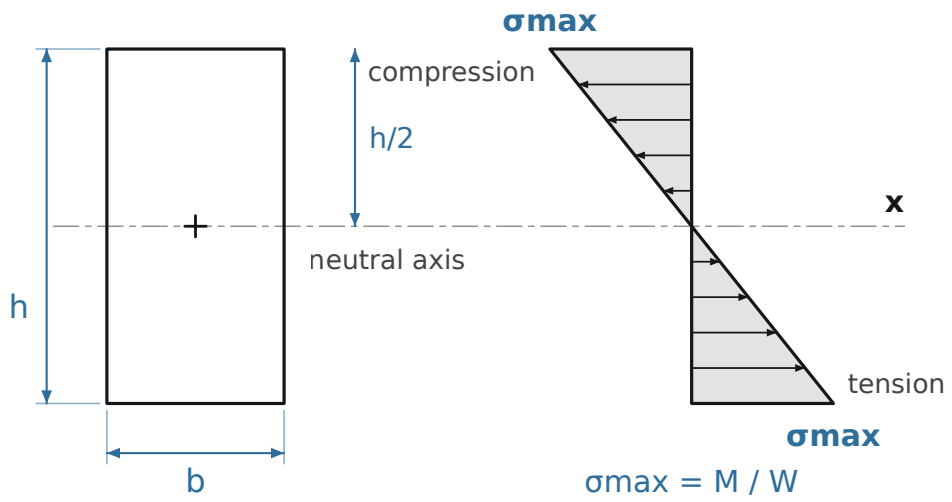
Area moment of inertia: section formulas, examples and units

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2. Formulas for common sections

Moments of inertia about the centroidal x axis (through the centroid), bending acting in the direction of h, H or D. W = I / c, c being the distance from the axis to the extreme fibre. J: polar moment of inertia, for the torsion of round sections.

Section	Area A	Moment of inertia I_x	Section modulus W_x	Polar J
Solid rectangle b × h	$b \cdot h$	$b \cdot h^3 / 12$	$b \cdot h^2 / 6$	—
Square, side a	a^2	$a^4 / 12$	$a^3 / 6$	—
Rectangular tube B × H, inside b × h	$B \cdot H - b \cdot h$	$(B \cdot H^3 - b \cdot h^3) / 12$	I / (H / 2)	—
Solid round, diameter D	$\pi \cdot D^2 / 4$	$\pi \cdot D^4 / 64$	$\pi \cdot D^3 / 32$	$\pi \cdot D^4 / 32$
Round tube D × d	$\pi \cdot (D^2 - d^2) / 4$	$\pi \cdot (D^4 - d^4) / 64$	$\pi \cdot (D^4 - d^4) / (32 \cdot D)$	$\pi \cdot (D^4 - d^4) / 32$
Symmetric I-section B × H, web t_w, flanges t_f	$2 \cdot B \cdot t_f + (H - 2 \cdot t_f) \cdot t_w$	$(B \cdot H^3 - (B - t_w) \cdot (H - 2 \cdot t_f)^3) / 12$	I / (H / 2)	—
Triangle, base b, height h	$b \cdot h / 2$	$b \cdot h^3 / 36$	$b \cdot h^2 / 24$ (apex)	—



3. Worked examples

Dimensions in millimetres. The same 20 × 40 rectangle is four times stiffer on edge than flat: the moment of inertia varies with the cube of the depth.

Section	A (mm^2)	I_x (cm^4)	W_x (cm^3)
Rectangle 20 × 40 on edge (h = 40)	800.0	**10.67**	5.33
The same, flat (h = 20)	800.0	**2.67**	2.67
Solid round Ø 20	314.2	**0.79**	0.79
Round tube Ø 40 × 2	238.8	**4.32**	2.16
Square tube 40 × 40 × 2	304.0	**7.34**	3.67

Section	A (mm^2)	I_x (cm^4)	W_x (cm^3)
I-section 100 × 200, web 5.6, flanges 8.5 (no fillets)	2 724.8	**1 845.59**	184.56

Rolled profiles (IPE, HEA…) have root fillets between web and flanges, which the formula ignores: their actual moment of inertia is slightly higher. Use the manufacturer's catalogue value.

5. Units and conversions

The moment of inertia is expressed in mm⁴ or cm⁴ (profile catalogues), the section modulus in mm³ or cm³. Exact conversions, 1 inch being 25.4 mm:

Unit	Equals
1 cm^4	10 000 mm^4
1 in^4	416 231.4 mm^4 = 41.62 cm^4
1 cm^3	1 000 mm^3
1 in^3	16 387.064 mm^3 = 16.39 cm^3

Reference values taken from the article https://www.baouch.fr/en/blog/moment-quadratique-inertie-section-calcul/, which explains the assumptions, formulas and edge cases. For critical use, check the current standard or the manufacturer's documentation. Free to print and share with credit to the source.